

Project Title	BrickView: Real Estate Analytics Platform
Skills take away From This Project	Python, SQL, Streamlit, Data Analysis
Domain	Real Estate, Property Analytics, Urban Development, Location Intelligence

Problem Statement

The real estate market is vast and dynamic, with properties being listed, sold, and evaluated every day. Buyers, sellers, and agents often lack accessible tools to monitor trends, pricing, and sales performance. This project aims to build a **Real Estate Listings Dashboard** that uses **SQL and Streamlit** to:

- Analyze property listings, agent performance, and sales patterns
- Provide insights into pricing, time on market, and property types
- Enable filtering by location, property type, price, and sales agent
- Display interactive visuals like maps and bar charts for better understanding.

Business Use Cases

- Assist buyers and investors in making data-informed decisions
- Help agents track sales performance and property types in demand



- Understand pricing trends across regions and neighborhoods
- Monitor time-on-market trends to improve sales strategies

Approach

1. Data Preparation

Use the provided datasets:

- Read raw JSON files using Python
- Flatten nested JSON structures (if any)
- Standardize date, numeric, and boolean fields
- Ensure date formats and price/area values are consistent

2. Database Creation

- Store data in SQL using normalized relationships
- Create views and indexes for performance

3. Data Analysis with SQL Queries

Use SQL to generate insights (detailed questions below)

4. Application Development with Streamlit

Create a user-friendly dashboard that allows:

- Filtering based on city, property type, agent, and price range
- Viewing maps of listings and bar/pie charts



- Displaying SQL query outputs as tables and visuals

5. Deployment

Deploy on a local or cloud server to allow real-time access by stakeholders

Data Flow and Architecture

Data Storage:

- SQL database with `listings`, `agents`, and `sales` etc. tables

Processing Pipeline:

- Use SQL for aggregation, joins, and trend analysis

Deployment:

- Streamlit UI for real-time insights and visualizations

Dataset:

1. `Listings` - [listings_final_expanded.json](#)
2. `Property_attributes` - [property_attributes_final_expanded.json](#)
3. `Agents` - [agents_cleaned.json](#)
4. `Sales` - [sales_cleaned.csv](#)
5. `buyers` - [buyers_cleaned.json](#)

Dataset Explanation:

- 1 listings – Property Listings

Core property-level information

Column	Description
Listing_ID	Unique ID for the property listing
City	City where the property is located
Property_Type	Apartment, Villa, Condo, etc.
Price	Listed price of the property
Area_sqft	Property size in square feet
Agent_ID	Foreign key to agents
Listed_Date	Date property was listed
Latitude	
Longitude	

2 property_attributes – Property Attributes

One-to-one with listings



Column	Description
Attribute_ID	Unique attribute record
Listing_ID	FK → listings
Bedrooms	Number of bedrooms
Bathrooms	Number of bathrooms
Floor_Number	Floor of the property
Total_Floors	Total floors in building
Year_Built	Year of construction
Is_Rented	Rented or not
Tenant_Count	Number of tenants
Furnishing_Status	Furnished / Semi / Unfurnished
Metro_Distance_Km	Distance to metro



Parking_Available	Parking availability
Power_Backup	Power backup availability

3 agents – Real Estate Agents

Column	Description
Agent_ID	Unique agent identifier
Name	Agent name
Phone	Phone number
Email	Email address
Commission_Rate	Commission %
Deals_Closed	Total deals
Rating	Client rating
Experience_Years	Years of experience



Avg_Closing_Days	Avg deal closing time
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5 sales – Property Sales

Column	Description
Listing_ID	FK → listings
Sale_Date	Sale date
Sale_Price	Final sale price
Days_On_Market	Time to sell

6 buyers – Buyer Information

Column	Description
Buyer_ID	Buyer identifier
Sale_ID	Sale id



Buyer_Type	Investor / End User
Payment_Mode	Cash / UPI / Bank / Cheque
Loan_Taken	Loan taken or not
Loan_Provider	Bank name
Loan_Amount	Loan amount

Key SQL Questions & Queries

Property & Pricing Analysis

1. What is the **average listing price by city**?
2. What is the **average price per square foot** by property type?
3. How does **furnishing status** impact property prices?
4. Do properties **closer to metro stations** command higher prices?
5. Are **rented properties** priced differently from non-rented ones?
6. How do **bedrooms and bathrooms** affect pricing?
7. Do properties with **parking and power backup** sell at higher prices?



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8. How does **year built** influence listing price?
 9. Which cities have the **highest average property prices**?
 10. How are properties distributed across **price buckets**?

Sales & Market Performance

11. What is the **average days on market** by city?
12. Which **property types sell the fastest**?
13. What percentage of properties are sold **above listing price**?
14. What is the **sale-to-list price ratio** by city?
15. Which listings took **more than 90 days** to sell?
16. How does **metro distance** affect time on market?
17. What is the **monthly sales trend**?
18. Which properties are currently **unsold**?

Agent Performance

19. Which agents have closed the **most sales**?
 20. Who are the **top agents by total sales revenue**?
 21. Which agents close deals **fastest**?
22. Does **experience correlate with deals closed**?
 23. Do agents with **higher ratings** close deals faster?



24. What is the **average commission earned** by each agent?

25. Which agents currently have the **most active listings**?



Buyer & Financing Behavior

26. What percentage of buyers are **investors vs end users**?

27. Which cities have the **highest loan uptake rate**?

28. What is the **average loan amount** by buyer type?

29. Which **payment mode** is most commonly used?

30. Do **loan-backed purchases** take longer to close?



Streamlit App Features



Filters Page

- **City** – Multi-select (e.g., filter listings in New York, San Francisco, etc.)
 - **Property Type** – Dropdown (Apartment, Villa, Condo, etc.)
 - **Price Range** – Slider for min and max price
 - **Agent** – Searchable dropdown to filter by agent
 - **Date Range** – Date picker for Listed Date or Sale Date
- etc.



Visualizations Page



- **Map:** Interactive map of current property listings by city
- **Bar Chart:** Number of listings or average prices by city
- **Pie Chart:** Distribution of property types
- **Line Chart:** Monthly sales and listings trend
- **Table View:** SQL query results with pagination and sorting

etc.

3 **CRUD Operations Page**

- Implement complete **CRUD (Create, Read, Update, Delete)** operations.
- Apply CRUD functionality to **all database tables**.
- Each table must support:
 - **View** records
 - **Add** new records
 - **Update** existing records
 - **Delete** records

4 **SQL Queries Display Page**

- Show all SQL queries in **drop-down format**.
- Each drop-down must include:
 - The **SQL query**
 - The **output displayed as a table**.



Results

- ✓ A full-featured Streamlit app to explore real estate data
- ✓ 15+ SQL queries providing insights into price, agent performance, and property types
- ✓ Visualizations and filters for interactive data exploration
- ✓ Clean database schema optimized for real-time querying

Project Evaluation Metrics:

- Proper SQL schema and normalized tables
- Accuracy of SQL queries and aggregations



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- Quality and completeness of Streamlit visualizations
 - Functional filters and interactivity
 - User-friendly interface and navigation

Technical Tags:
Python, SQL, Streamlit, Real Estate Analytics, Visualization

- Deliverables:**
- ✓ Cleaned CSV data and SQL schema
 - ✓ 15+ SQL queries for insights and KPIs
 - ✓ Streamlit app with filtering, visuals, and agent/property dashboards
 - ✓ Final presentation/report with insights and screenshots

- Timeline:**
- 1 Weeks

References:

Project Live Evaluation Metrics	Project Live Evaluation
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EDA Guide	 Exploratory Data Analysis (EDA) Guide
Capstone Explanation Guideline	 Capstone Explanation Guideline
GitHub Reference	 How to Use GitHub.pptx
Streamlit recordings (English)	 Special session for STREAMLIT(11/...
STREAMLIT RECORDING (Tamil)	 Streamlit - Tamil
Streamlit documentation	Install Streamlit  Streamlit App on Streamlit Cloud str_app.py
HOW TO ESTABLISH SQL CONNECTION:	PYTHONSQLCODE_TAMIL.ipynb
Project orientation (English)	https://classify.zenclass.in/meet-video?session=f6f452f9-d947-4ea9-ad46-06507eb55b44&index=0 DS-C-WE-E-B147 & DS-C-WE-E-B148: https://classify.zenclass.in/meet-video?session=9b563eb5-662e-449c-87ed-c3c5a88c72ad&index=0
	DS-C-WE-E-B133 & DS-C-WE-E-B134: https://classify.zenclass.in/meet-video?session=e51eadd0-8518-4f38-bd90-5d9de59f85a9&index=0



Project orientation (Hindi)	https://classify.zenclass.in/meet-video?session=aef1ede0-609b-4d59-b2be-a7cb519b7aee&index=0  Project Orientation Session Link: D...
Project Excellence Series (English)	 Project Excellence Series: Guided Le...
Project Excellence Series (Tamil)	 Project Excellence Series: Guided Le...
Project Orientation (English)(reference)	 Project Orientation Session : BankSi...
Project Orientation (Hindi)(ref)	 Project Orientation Session : BankSi...
Project Orientation (Tamil)(ref)	https://classify.zenclass.in/meet-video?session=95c8ec76-4667-485c-9102-747d00d108e5&index=0
Demo code	sample (1).ipynb
Study materials	 python_study_material_final.pdf  SQL_final.pdf
How to install vscode along with python	 How_to_install_vscode_along_with_...

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Note: Book the slot at least before 12:00 Pm on the same day



Timing: Monday-Friday (3:00PM to 4:00PM)

Saturday (2:00 PM to 3:00 PM)

Session link :

 **Project Doubt Session [DS/AIML]**

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:

For DS and AIML

Monday-Thursday (11:00AM to 12:00 Noon) and (05:00PM to 06:00PM)

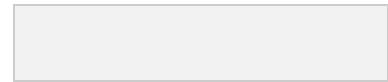
Saturday (11:00AM to 12:00PM)

Booking link : <https://forms.gle/1m2Gsro41fLtZurRA>

Evaluation Metrics : [Project Live Evaluation](#)

Project Created By	Verified By	Approved By
Vinodhini	Nehlath Harmain	Nehlath Harmain





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